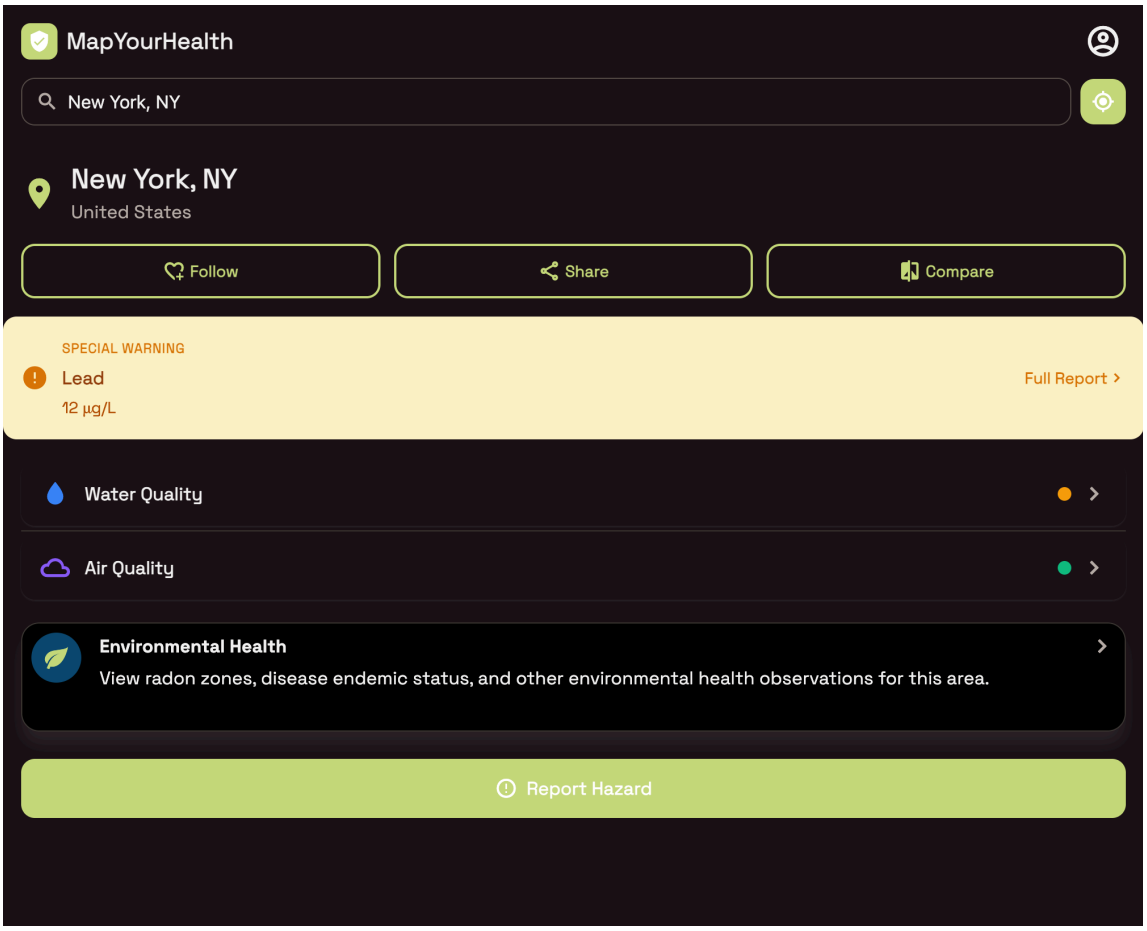
How Your Safety Data Gets to Your Screen

This guide shows exactly how data from our source spreadsheet appears on your phone.

Screenshots from the App

1. Location Overview (City Search)



When you search for a **city name** (e.g., "New York, NY"), the app shows:

- **Location header** - Your searched city (New York, NY)
- **Warning banner** - Highlights the most critical issue (Lead at 12 µg/L)
- **Category cards** - Water Quality, Air Quality, etc.

2. Location Overview (Address Search)

MapYourHealth

New York, NY

New York, NY

United States

Showing data for nearest city to: 350 Fifth Avenue, New York, NY, USA

Follow

Share

Compare

SPECIAL WARNING

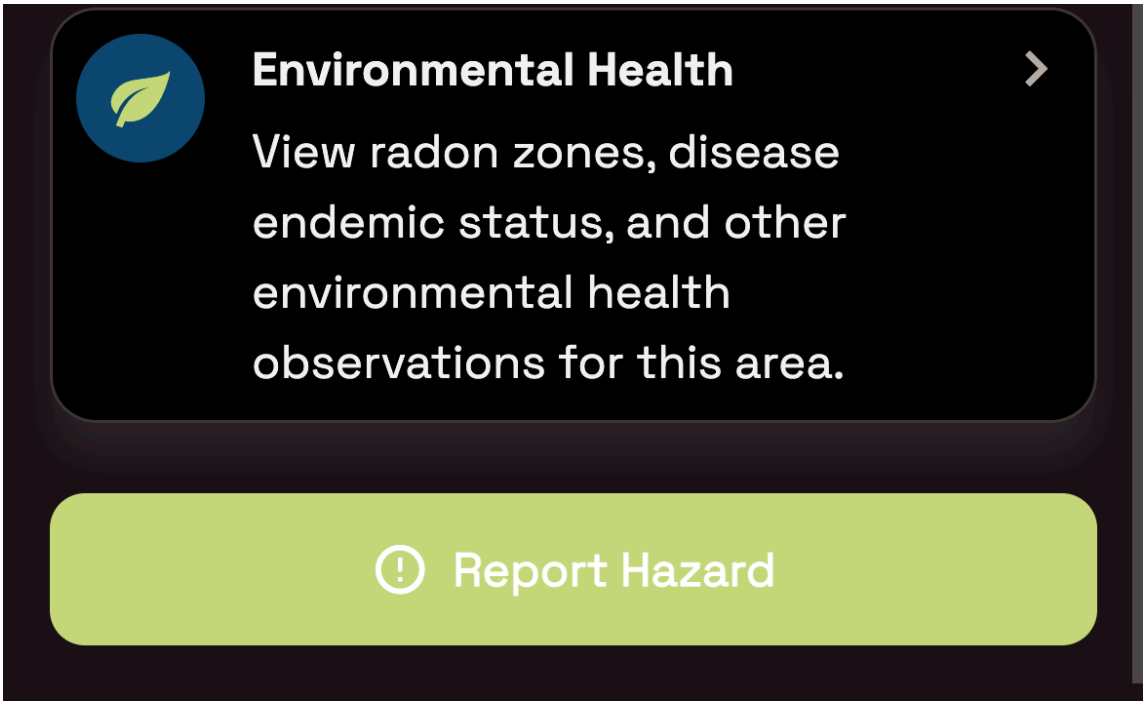
Lead

12 µg/L

Full Report >

Water Quality

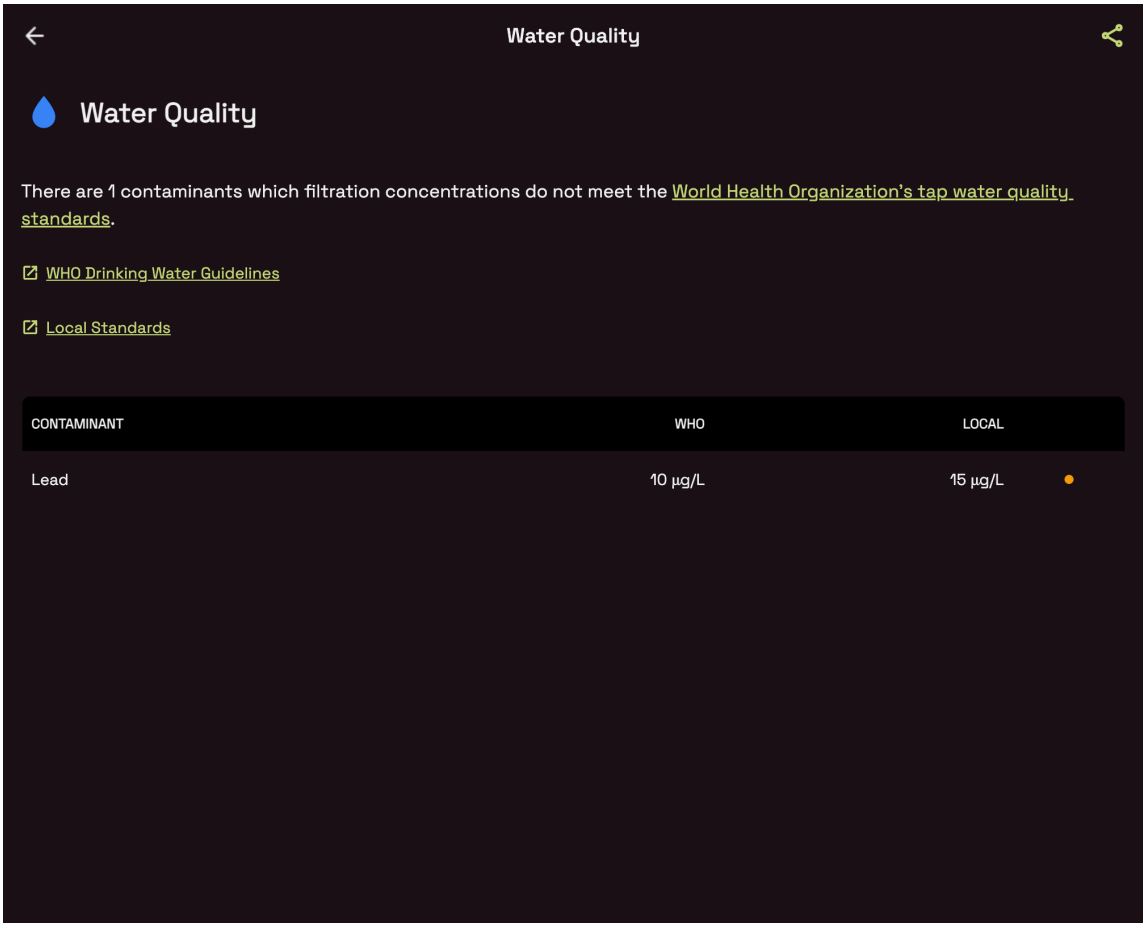
Air Quality



When you search for a **street address** (e.g., "350 Fifth Avenue, New York, NY, USA"), the app:

- Finds the **nearest city** in our database to that address
- Shows: *"Showing data for nearest city to: 350 Fifth Avenue, New York, NY, USA"*
- Displays the same data as a city search, using the resolved city's jurisdiction

3. Category Detail View



The detail view shows:

- **Category header** - Water Quality with icon
- **Description** - Dynamic count of contaminants exceeding limits
- **External links** - WHO guidelines, Local standards
- **Contaminant table** - Side-by-side comparison of WHO vs Local limits

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The Source: Risks.xlsx

Our water quality data comes from a master spreadsheet with 172+ contaminants:

A	B	C	D	E	...	K
NAME	WHO	QUEBEC	NY	CA		DESCRIPTION_EN
Lead	10	10	15	10		Neurotoxin...
Nitrate	50	10	10	10		Blue baby...
Arsenic	10	10	10	10		Carcinogen...

Field Mapping: Spreadsheet → App Screen

Here's exactly how each spreadsheet column becomes a UI element on the detail screen:

The Contaminant Table

Looking at the screenshot, here's how the data maps:

Spreadsheet Column	Table Column	Example Value
Column A (NAME)	CONTAMINANT	"Lead"
Column B (WHO)	WHO	"10 µg/L"
Column D (NY)	LOCAL	"15 µg/L"
Calculated	Status dot	🟡 (yellow = warning)

Visual Mapping

```
SPREADSHEET (Risks.xlsx)                    APP SCREEN
──────────────────────────────────────────────────────────────────────


Column A: "Lead"                    →       CONTAMINANT: "Lead"

Column B: 10                        →       WHO: "10 µg/L"
                                            (We add the unit automatically)


Column D: 15                        →       LOCAL: "15 µg/L"
(NY column, based on your location)         (Shows YOUR state's limit)

Measured value: 12 µg/L             →       Status: 🟡
(from LocationMeasurement table)            (12 is 80% of 15 = warning)
```

How Your Address Becomes a Jurisdiction

When you type an address or city name, the app resolves it to a regulatory jurisdiction in three steps:

Step 1: Find Your City

```
You type: "350 Fifth Avenue, New York"
                    │
                    ▼
        ┌──────────────────────────────┐
        │  Search Local Database       │ ← We check our Location table first
        │  (172+ cities indexed)       │
        └──────────────────────────────┘
                    │
                    │
             Found match? ─── Yes ──→ Use city/state from database
                    │
                    No
```

```
                  │
                  ▼
┌──────────────────────────────┐
│  Google Places API           │ ← Fallback for addresses not in DB
│  (US & Canada only)          │
└──────────────────────────────┘
                  │
                  ▼
┌──────────────────────────────┐
│  Resolve Coordinates         │ ← Get lat/lng from the address
│  Find Nearest City           │ ← Match to closest city we have
│  in our Location table       │    data for
└──────────────────────────────┘
```

Result: `city: "New York"` , `state: "NY"` , `country: "US"`

This is why the app shows *"Showing data for nearest city to: 350 Fifth Avenue, New York, NY, USA"* when you search by address (see Screenshot 2 above).

Step 2: Map State to Jurisdiction Code

Your state or province is mapped to a jurisdiction code:

Your State/Province	Jurisdiction Code	Meaning
New York	US-NY	New York state regulations
California	US-CA	California state regulations
Texas	US-TX	Texas state regulations
Florida	US-FL	Florida state regulations
Quebec	CA-QC	Quebec provincial regulations
Ontario	CA-ON	Ontario provincial regulations

If your state doesn't have specific regulations, we fall back:

1. **State-specific** (e.g., `US-NY`) — first choice
2. **Country federal** (e.g., `US`) — if no state-level data exists
3. **WHO** — global default

Step 3: Look Up Safety Limits (Thresholds)

The jurisdiction code determines which "LOCAL" column you see:

Jurisdiction Code	Spreadsheet Column	What Shows as "LOCAL"
US-NY	Column D (NY)	NY state limits
US-CA	Column E (CA)	CA state limits
US-TX	Column F (TX)	TX state limits
US-FL	Column G (FL)	FL state limits

CA-QC	Column C (QUEBEC)	Quebec limits

What Is a Threshold?

A **threshold** is the maximum safe level of a contaminant allowed by a specific jurisdiction. Each threshold record in our database contains:

Field	Description	Example
Contaminant	Which substance	Lead
Jurisdiction	Who sets the limit	US-NY (New York)
Limit Value	Maximum allowed concentration	15 µg/L
Warning Ratio	% of limit that triggers a warning	0.80 (80%)
Status	Regulatory status	regulated

Why Thresholds Matter

Different jurisdictions set different limits for the same contaminant. For example, Lead:

Jurisdiction	Limit	Meaning
WHO	10 µg/L	International guideline
US-NY (New York)	15 µg/L	NY state law allows more
US-CA (California)	10 µg/L	CA matches WHO

The app always shows **two thresholds** side by side — WHO (international) and LOCAL (your jurisdiction) — so you can compare how your area's regulations stack up against global health recommendations.

Threshold Statuses

Not every contaminant has a numeric limit. A threshold can have one of these statuses:

Status	What It Means	How It Appears in the App
Regulated	Has a numeric limit (e.g., 15 µg/L)	`15 µg/L`
Banned	Completely prohibited — no amount is allowed	`BANNED`
Not Approved	Not approved for use in this jurisdiction	`NOT APPROVED`
Not Controlled	No standard has been set	`NO STANDARD`

The Warning Ratio

Each threshold has a **warning ratio** (default: 0.80 or 80%). This determines when the app shows a warning *before* a contaminant reaches its legal limit.

```
Example: Lead in New York (limit = 15 µg/L, warning ratio = 0.80)

Warning threshold = 15 × 0.80 = 12 µg/L

  0 µg/L          12 µg/L          15 µg/L
  |—— 🟢 SAFE ——|—— 🟡 WARNING —|—— 🔴 DANGER ——→
```

This gives you an early heads-up when contaminant levels are approaching the legal limit, even if they haven't exceeded it yet.

Status Colors Explained

The colored dot indicates risk level, based on the threshold and warning ratio:

Color	Meaning	Calculation
🔴 Red	**DANGER**	Measured value ≥ Local limit
🟡 Yellow	**WARNING**	Measured value ≥ (limit × warning ratio)
🟢 Green	**SAFE**	Measured value < (limit × warning ratio)

Example: Lead in New York

From the screenshot:

- **WHO limit**: 10 µg/L (from Column B)
- **Local limit**: 15 µg/L (from Column D - NY)
- **Measured value**: 12 µg/L (from your water report)

Calculation:

```
Warning threshold = 15 × 0.80 = 12 µg/L

Is 12 ≥ 15?  → No  → Not danger
Is 12 ≥ 12?  → Yes → WARNING (🟡)
```

Special Values in the Spreadsheet

Spreadsheet Value	What It Means	How It Appears
`10` (number)	Regulated at 10 µg/L	`10 µg/L`
`BANNED`	Completely prohibited	`BANNED`
`NOT APPROVED`	Not approved for use	`NOT APPROVED`
(empty cell)	No standard set	`NO STANDARD`

Why Two Standards?

We show both **WHO** and **LOCAL** so you can see:

1. **WHO** - The international health benchmark
2. **LOCAL** - What your state/province legally requires

In the Lead example:

- WHO says: Keep it under **10 µg/L**
- New York says: Keep it under **15 µg/L**

The WHO standard is stricter. Your water (12 µg/L) meets NY requirements but exceeds WHO recommendations - hence the warning.

Complete Data Flow

```
┌──────────────────────────────────────────────────────────────────────────┐
│                        RISKS.XLSX (Source)                               │
│                                                                          │
│    Lead   │  10  │  10  │  15  │  10  │  ...  │  Neurotoxin affecting...  │
│           │ WHO  │  QC  │  NY  │  CA  │       │  (description)            │
└──────────────────────────────────────────────────────────────────────────┘
                                  │
                                  ▼
                     ┌──────────────────────────┐
                     │       PARSING SCRIPT     │
                     │  • Extract: "Lead"       │
                     │  • WHO limit: 10         │
                     │  • US-NY limit: 15       │
                     │  • Unit: µg/L            │
                     └──────────────────────────┘
                                  │
                                  ▼
                     ┌──────────────────────────┐
                     │         DATABASE         │
                     │  Contaminant: lead       │
                     │  ContaminantThreshold:   │
                     │    WHO: 10 µg/L          │
                     │    US-NY: 15 µg/L        │
                     └──────────────────────────┘
                                  │
                                  ▼
                     ┌──────────────────────────┐
                     │    LOCATION MEASUREMENT  │
                     │  City: New York          │
                     │  State: NY               │
                     │  Lead: 12 µg/L           │
                     │  Source: Water utility report │
                     └──────────────────────────┘
                                  │
                                  ▼
┌──────────────────────────────────────────────────────────────────────────┐
│                        APP SCREEN (What You See)                         │
```

```
│                                                                          │
│  ┌──────────────────────────────────────────────────────────────────┐   │
│  │ CONTAMINANT     │    WHO        │   LOCAL       │  STATUS          │   │
│  ├─────────────────┼───────────────┼───────────────┼──────────────────┤   │
│  │ Lead            │  10 µg/L      │  15 µg/L      │    🟡            │   │
│  └─────────────────┴───────────────┴───────────────┴──────────────────┘   │
│                                                                          │
│  Status is WARNING because 12 µg/L is ≥ 80% of the 15 µg/L limit          │
└──────────────────────────────────────────────────────────────────────────┘
```

Questions?

If you notice discrepancies between our data and official reports, please let us know. We continuously update our database as regulations change.

If you'd like to add other data sources, consume data from an external API, or notice anything missing from this guide, please let us know — we're happy to integrate new sources and keep this document up to date.